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Window hinge assembly and methods

Abstract

Window hinge assemblies including a hinge pin support and methods of using the same are described herein. The hinge assemblies include a track assembly, a shoe slidably engaged in a track of the track assembly, a sash arm configured for attachment to a rotating sash, the sash arm being pivotally attached to the shoe, and a connecting arm. The track assembly includes a base extending along a track axis, a shoe track extending along the track axis, a hinge pin attached to the base of the track assembly, and a hinge pin support restraining an upper end of the hinge pin from movement relative to the base.

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Background/Summary

RELATED APPLICATONS (1) This is a continuation application of U.S. patent application Ser. No. 16/858,101, filed Apr. 24, 2020, which claims the benefit of U.S. Provisional Application No. 62/878,954 entitled “WINDOW HINGE ASSEMBLY AND METHODS” and filed on Jul. 26, 2019 and also claims the benefit of U.S. Provisional Patent Application Ser. No. 62/882,766 entitled “IMPROVED WINDOW HINGE ASSEMBLY AND METHODS” and filed on Aug. 5, 2019, both of which are incorporated herein by reference in their entireties. (2) Window hinge assemblies including a hinge pin support and methods of using the same are described herein.

BACKGROUND

(1) Casement windows include a frame mounted in a rough opening provided in a structure, with at least one sash mounted for rotation within the window frame, the sash being movable between open and closed positions within the window frame.

(2) The sashes in such windows are mounted on hinge assemblies to allow for rotation of the sash within the window frame. Those hinge assemblies may include a track attached to the window frame and a sash arm attached to the sash. One end of the sash arm is attached to a shoe that slides along the track, with the sash arm being pivotally attached to the shoe. The hinge assembly further includes a connecting arm pivotally attached to both the track and the sash arm. As the sash rotates from a closed position to an open position, the shoe slides along the track while the connecting arm rotates around a hinge pin positioned in the window frame. Some examples of such known hinge assemblies for casement windows may be found in, for example, U.S. Pat. Nos. 5,491,930; 6,112,371; 6,134,751; 6,643,896; 7,735,194; and 8,468,656.

SUMMARY

(3) Window hinge assemblies including a hinge pin support and methods of using the same are described herein. In one or more embodiments, the hinge assemblies include a track assembly, a shoe slidably engaged in a track of the track assembly, a sash arm configured for attachment to a rotating sash, the sash arm being pivotally attached to the shoe, and a connecting arm. The track assembly includes a base extending along a track axis, a shoe track extending along the track axis, a hinge pin attached to the base of the track assembly, and a hinge pin support restraining an upper end of the hinge pin from movement relative to the base.

(4) One potential advantage of one or more embodiments of hinge assemblies as described herein is that restraining the upper end of the hinge pin in addition to supporting the base end of the hinge pin (where the hinge pin attaches to the base of the track assembly) can reduce the likelihood of damage to the hinge pin when the sash attached in the window frame is in a closed position. When the sash is in, for example, a closed position, forces acting on the window frame and directed along the track axis can be transmitted to the hinge pin through the connecting arm and the sash arm. Those forces can damage hinge pins that are supported only at their base end (where the hinge pins are attached only to the base of the track assembly-essentially resulting in a hinge pin cantilevered above the base of the track assembly).

(5) Restraining the upper end of the hinge pin in addition to supporting its base end with the sash in the closed position reduces the likelihood of damage to the hinge pin when, for example, the window unit is subjected to drops and other types of full window/sash impacts during transportation of the window unit from the assembly factory to the building rough opening into which it is installed. Restraining the upper end of the hinge pin in addition to supporting its base end can also reduce the likelihood of damage to a hinge pin when the sash is in an open position and subject to forces from, for example, wind gusts, which may be transmitted through the hinge assembly to the hinge pin, thereby damaging the hinge pin.

(6) Supporting the hinge pin at both its base end and its upper end distributes forces transmitted to the hinge pin through the connecting arm and can, in many instances, prevent or at least substantially reduce the likelihood of hinge pin damage.

(7) In a first aspect, one or more embodiments of a hinge assembly for supporting a rotating sash in

a window frame as described herein may include: a track assembly configured for attachment to a window frame, the track assembly comprising a base, a shoe track, a hinge pin, and a hinge pin support, wherein the base extends along a track axis from a hinge pin end to a shoe track end, wherein the shoe track extends along the track axis, and wherein the shoe track extends from the shoe track end towards the hinge pin end, wherein the hinge pin comprises a base end connected to the base proximate the hinge pin end of the base, the hinge pin comprising an upper end distal from the base, wherein the hinge pin support extends above the base such that a portion of the hinge pin is exposed between the base and the hinge pin support, wherein the hinge pin support restrains the upper end of the hinge pin from movement relative to the base. The hinge assembly may further include a shoe slidably engaged with the shoe track, wherein the shoe is configured to slide along the shoe track end the shoe is restrained from moving transverse to the track axis when the shoe is slidably engaged with the shoe track; a sash arm configured for attachment to a rotating sash, the sash arm extending from a shoe end to a distal end, wherein the shoe end is pivotally coupled to the shoe; and a connecting arm extending from a hinge pin end to a sash end, wherein the sash end is pivotally connected to the sash arm at an intermediate location spaced from the shoe end of the sash arm, and wherein the hinge pin end of the connecting arm is pivotally connected to the hinge pin for rotation about a pin axis extending through the upper end of the hinge pin, wherein the hinge pin end of the connecting arm is located between the hinge pin support and the base of the track assembly.
(8) In one or more embodiments of the hinge assemblies described herein, the hinge pin support comprises a hinge pin bore, wherein a portion of the hinge pin is located in the hinge pin bore such that movement of the hinge pin relative to the hinge pin support is restricted.
(9) In one or more embodiments of the hinge assemblies described herein, the hinge pin support comprises a hinge pin support tab formed integrally with the base, wherein the hinge pin support tab and the base define a U-shaped structure comprising a bottom from which two legs extend, wherein the hinge pin extends between the legs of the U-shaped structure distal from the bottom of the U-shaped structure.
(10) In one or more embodiments of the hinge assemblies described herein, the hinge pin support comprises a hinge pin support bracket separate and discrete from the base of the track assembly, wherein at least a portion of the hinge pin support bracket is positioned over the base such that the hinge pin extends between the base and the hinge pin support bracket. In one or more embodiments, the hinge pin support bracket and the base of the track assembly comprise interlocking mechanical features configured to transfer mechanical loads directed along the track axis on the hinge pin to the base and the hinge support bracket.
(11) In one or more embodiments of the hinge assemblies including a hinge pin support bracket as described herein, the hinge pin support bracket is secured relative to the base by a mechanical fastener. In one or more embodiments, the base comprises a base fastener aperture and wherein the hinge pin support bracket comprises a bracket fastener aperture, wherein the base fastener aperture is aligned with the bracket fastener aperture when the hinge pin support bracket is secured relative to the base by the mechanical fastener. In one or more embodiments, the hinge pin support bracket and the base of the track assembly comprise interlocking mechanical features configured to transfer mechanical loads directed along the track axis on the hinge pin to the base and the hinge pin support bracket when the hinge pin support bracket is secured relative to the base by the mechanical fastener.
(12) In one or more embodiments of the hinge assemblies including a hinge pin support bracket and base including interlocking mechanical features as described herein, the interlocking mechanical features are located proximate the base fastener opening and the bracket fastener opening.
(13) In one or more embodiments of the hinge assemblies including a hinge pin support bracket and base including interlocking mechanical features as described herein, the interlocking

mechanical features are positioned around a perimeter of the base fastener opening and around a perimeter of the bracket fastener opening.
(14) In one or more embodiments of the hinge assemblies including a hinge pin support bracket as described herein, the bracket fastener opening comprises an elongated non-circular opening such that a position of the hinge pin bracket relative to the base is adjustable along the track axis.
(15) In one or more embodiments of the hinge assemblies including a hinge pin support bracket as described herein, the hinge pin support bracket further comprises a vent limiter configured to limit rotation of the connecting arm around the hinge pin when the hinge assembly is moving from a closed configuration to an open configuration. In one or more embodiments, the hinge pin support bracket and the vent limiter each comprise separate and discrete articles attached to each other. In one or more embodiments, the hinge pin support bracket and the vent limiter are attached to each other using one or more mechanical fasteners.
(16) In one or more embodiments of the hinge assemblies described herein, the hinge assembly further comprises a bushing in the hinge pin bore, wherein the hinge pin comprises a pin portion located in a bushing aperture of the bushing, wherein the bushing aperture and the pin portion of the hinge pin are positioned off-center in the hinge pin bore such that rotation of the hinge pin about a stud axis extending through a center of the hinge pin bore rotates the bushing in the hinge pin bore and moves the pin portion of the hinge pin towards or away from the shoe track. In one or more embodiments, the bushing and the hinge pin bracket comprise visible indicia indicative of the rotational position of the bushing in the hinge pin bore. In one or more embodiments, the relative rotational position of the bushing is indicative of the position of the pin portion of the hinge pin along the track axis relative to the shoe track.
(17) In a second aspect, one or more embodiments of a hinge assembly for supporting a rotating sash in a window frame as described herein may include: a track assembly configured for attachment to a window frame, the track assembly comprising a base, a shoe track, a hinge pin, and a hinge pin support, wherein the base extends along a track axis from a hinge pin end to a shoe track end, wherein the shoe track extends along the track axis, and wherein the shoe track extends from the shoe track end towards the hinge pin end, wherein the hinge pin comprises a base end connected to the base proximate the hinge pin end of the base, the hinge pin comprising an upper end distal from the base, wherein the hinge pin support comprises a hinge pin support bracket separate and discrete from the base of the track assembly, wherein the hinge pin support bracket comprises a support portion extending above the base, the support portion comprising a hinge pin bore, wherein a portion of the hinge pin is exposed and extends between the base and the hinge pin bore of the support portion of the hinge pin support, wherein the hinge pin support restrains the upper end of the hinge pin from movement relative to the base, and wherein the hinge pin support bracket and the base of the track assembly comprise interlocking mechanical features configured to transfer mechanical loads directed along the track axis on the hinge pin to the base and the hinge pin support bracket when the hinge pin support bracket is secured relative to the base. The hinge assemblies may further include a shoe slidably engaged with the shoe track, wherein the shoe is configured to slide along the shoe track end the shoe is restrained from moving transverse to the track axis when the shoe is slidably engaged with the shoe track; a sash arm configured for attachment to a rotating sash, the sash arm extending from a shoe end to a distal end, wherein the shoe end is pivotally coupled to the shoe; and a connecting arm extending from a hinge pin end to a sash end, wherein the sash end is pivotally connected to the sash arm at an intermediate location spaced from the shoe end of the sash arm, and wherein the hinge pin end of the connecting arm is pivotally connected to the hinge pin for rotation about a pin axis extending through the upper end of the hinge pin, wherein the hinge pin end of the connecting arm is located between the hinge pin support and the base of the track assembly. In one or more embodiments, the hinge pin support bracket further comprises a vent limiter configured to limit rotation of the connecting arm around the hinge pin when the hinge assembly is moving from a closed configuration to an open

configuration.

(18) In a third aspect, one or more embodiments of a hinge assembly for supporting a rotating sash in a window frame as described herein includes: a track assembly configured for attachment to a window frame, the track assembly comprising a base, a shoe track, a hinge pin, a hinge pin support bracket, and a bracket clip, wherein the base extends along a track axis from a hinge pin end to a shoe track end, wherein the shoe track extends along the track axis, and wherein the shoe track extends from the shoe track end towards the hinge pin end, wherein the hinge pin comprises a base end connected to the base proximate the hinge pin end of the base, the hinge pin comprising an upper end distal from the base, wherein the hinge pin support bracket is separate and discrete from the base of the track assembly, wherein the hinge pin support bracket comprises a support portion extending above the base and a base portion located above the base of the track assembly, wherein the support portion of the hinge pin support bracket comprises a hinge pin bore, wherein a portion of the hinge pin is exposed and extends between the base and the hinge pin bore of the support portion of the hinge pin support bracket, wherein the bracket clip is positioned above the base portion of the of the hinge pin support bracket such that the base portion of the hinge pin support bracket is located between the bracket clip and the base of the track assembly, wherein the hinge pin support bracket and the bracket clip are configured to restrain the upper end of the hinge pin from movement relative to the base, wherein the bracket clip, the base portion of the hinge pin support bracket and the base of the track assembly comprise interlocking mechanical features configured to transfer at least a portion of any mechanical loads directed along the track axis on the hinge pin to the base through the hinge pin support bracket and the bracket clip when the hinge pin support bracket and the bracket clip are secured relative to the base and the upper end of the hinge pin is located in the hinge pin bore of the support portion of the hinge pin support bracket. The hinge assembly further includes a shoe slidably engaged with the shoe track, wherein the shoe is configured to slide along the shoe track end the shoe is restrained from moving transverse to the track axis when the shoe is slidably engaged with the shoe track; a sash arm configured for attachment to a rotating sash, the sash arm extending from a shoe end to a distal end, wherein the shoe end is pivotally coupled to the shoe; and a connecting arm extending from a hinge pin end to a sash end, wherein the sash end is pivotally connected to the sash arm at an intermediate location spaced from the shoe end of the sash arm, and wherein the hinge pin end of the connecting arm is pivotally connected to the hinge pin for rotation about a pin axis extending through the upper end of the hinge pin, wherein the hinge pin end of the connecting arm is located between the support portion of the hinge pin support bracket and the base of the track assembly.

(19) In a fourth aspect, one or more embodiments of a hinge assembly for supporting a rotating sash in a window frame as described herein may include: a track assembly configured for attachment to a window frame, the track assembly comprising a base, a shoe track, a hinge pin, and a hinge pin support, wherein the base extends along a track axis from a hinge pin end to a shoe track end, wherein the shoe track extends along the track axis, and wherein the shoe track extends from the shoe track end towards the hinge pin end, wherein the hinge pin comprises a base end connected to the base proximate the hinge pin end of the base, the hinge pin comprising an upper end distal from the base, wherein the hinge pin support extends above the base such that a portion of the hinge pin is exposed between the base and the hinge pin support, wherein the hinge pin support restrains the upper end of the hinge pin from movement relative to the base, wherein the hinge pin support comprises a hinge pin bore, wherein a portion of the hinge pin is located in the hinge pin bore such that movement of the hinge pin relative to the hinge pin support is restricted, and wherein the hinge pin support comprises a hinge pin support tab formed integrally with the base, the hinge pin bore being located in the hinge pin support tab, wherein the hinge pin support tab and the base define a U-shaped structure comprising a bottom from which two legs extend, wherein the hinge pin extends between the legs of the U-shaped structure distal from the bottom of the U-shaped structure. The hinge assemblies may further include a shoe slidably engaged with the

shoe track, wherein the shoe is configured to slide along the shoe track end the shoe is restrained from moving transverse to the track axis when the shoe is slidably engaged with the shoe track; a sash arm configured for attachment to a rotating sash, the sash arm extending from a shoe end to a distal end, wherein the shoe end is pivotally coupled to the shoe; and a connecting arm extending from a hinge pin end to a sash end, wherein the sash end is pivotally connected to the sash arm at an intermediate location spaced from the shoe end of the sash arm, and wherein the hinge pin end of the connecting arm is pivotally connected to the hinge pin for rotation about a pin axis extending through the upper end of the hinge pin, wherein the hinge pin end of the connecting arm is located between the hinge pin support and the base of the track assembly.

(20) As used herein and in the appended claims, the singular forms "a," "an," and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a" or "the" component may include one or more of the components and equivalents thereof known to those skilled in the art. Further, the term "and/or" means one or all of the listed elements or a combination of any two or more of the listed elements.

(21) It is noted that the term "comprises" and variations thereof do not have a limiting meaning where these terms appear in the accompanying description. Moreover, "a," "an," "the," "at least one," and "one or more" are used interchangeably herein.

(22) The above summary is not intended to describe each embodiment or every implementation of the hinge assemblies and methods of using the same as described herein. Rather, a more complete understanding of the invention will become apparent and appreciated by reference to the following Description of Illustrative Embodiments and claims in view of the accompanying figures of the drawing.

Description

BRIEF DESCRIPTION OF THE VIEWS OF THE DRAWING

(1) FIG. **1** is a perspective view of one illustrative embodiment of a hinged window incorporating one illustrative embodiment of a hinge assembly as described herein (portions of the sash being removed to facilitate viewing of the hinge assembly and the window frame).

(2) FIG. **2**A is a perspective view depicting the hinge assembly of FIG. **1** in isolation from the window frame and sash depicted in FIG. **1**.

(3) FIG. **2**B is a top view of the components of the hinge assembly of FIG. **2**A with the sash in the closed position such that the sash arm **50** and connecting arm **60** are aligned along the track axis **21**.

(4) FIG. **2**C is an enlarged cross-sectional view of the hinge assembly of FIG. **2**B taken along line **2**C-**2**C in FIG. **2**B.

(5) FIG. **3** is a top view of the illustrative embodiment of a hinge pin support bracket as depicted in FIGS. **1-2**.

(6) FIG. **4** is a side elevation view of the hinge pin support bracket of FIG. **3**.

(7) FIG. **5**A is an enlarged top view of the hinge pin support bracket located on the base of the track assembly.

(8) FIG. **5**B is a cross-sectional view of FIG. **5**A taken along line **5**B-**5**B in FIG. **5**A.

(9) FIG. **6**A is a view of the hinge pin support bracket of FIG. **5**A after rotation of the hinge pin relative to the track and depicting corresponding movement of the hinge pin bracket.

(10) FIG. **6**B is a cross-sectional view of FIG. **6**A taken along line **6**B-**6**B in FIG. **6**A.

(11) FIG. **7** it is a top view of another illustrative embodiment of a hinge pin support bracket that may be used in one or more embodiments of a hinge assembly as described herein.

(12) FIG. **8** is an enlarged cross-sectional view of the hinge pin support bracket of FIG. **7** taken along line **8-8** in FIG. **7**.

(13) FIG. **9** is a bottom view of the hinge pin support bracket of FIGS. **7-8**.
(14) FIG. **10** is a perspective view of another illustrative embodiment of a hinge pin support bracket that may be used in one or more embodiments of a hinge assembly as described herein.
(15) FIG. **11** is a top view of the hinge pin support bracket of FIG. **10** in position to support a hinge pin of a track assembly of a hinge assembly as described herein.
(16) FIG. **12** is a top view of a portion of another illustrative embodiment of a track assembly including a hinge pin support as described herein.
(17) FIG. **13** is a side view of the track assembly depicted in FIG. **12**.
(18) FIG. **14** is a view of the track assembly of FIG. **13** with the mechanical fastener and bushing removed.
(19) FIG. **15** it is a top view of another illustrative embodiment of a hinge pin support bracket that may be used in one or more embodiments of a hinge assembly as described herein.
(20) FIG. **16** is a cross-sectional view of the hinge pin support bracket of FIG. **15** taken along line **16-16** in FIG. **15**.
(21) FIG. **17** is a top view of the hinge pin support bushing of the hinge pin support bracket of FIG. **15** removed from the hinge pin support bracket.
(22) FIG. **18** is a cross-sectional view of the hinge pin support bushing of FIG. **17** taken along line **18-18** in FIG. **17**.
(23) FIG. **19** is a top view of another illustrative embodiment of a hinge pin support bracket including one illustrative embodiment of a vent limiter that may be used in in one of more embodiments of a hinge assembly as described herein.
(24) FIG. **20** depicts the hinge pin support bracket and vent limiter of FIG. **20** on a portion of a hinge assembly, with the vent limiter limiting rotation of a connecting arm around the hinge pin.
(25) FIG. **21** is a side view of another illustrative embodiment of a hinge pin support bracket including an alternative hinge pin one that may be used in in one of more embodiments of a hinge assembly as described herein.
(26) FIG. **22** is a perspective view of another illustrative embodiment of the hinge pin end of a hinge assembly as described herein.
(27) FIG. **23** is an exploded perspective view of the hinge pin support bracket, bracket clip, and fastener of the hinge pin assembly of FIG. **22**.
(28) FIG. **24** is a top plan view of hinge pin support bracket, bracket clip, and fastener of the hinge pin assembly of FIG. **23**.
(29) FIG. **25** is a cross-sectional view of the hinge pin support bracket, bracket clip, and fastener of the hinge pin assembly of FIGS. **23-24** taken along line **25-25** in FIG. **24**.
(30) FIG. **26** is a perspective view of the underside of the illustrative embodiment of the bracket clip depicted in FIGS. **21-25**.
(31) FIG. **27** is a plan view of the underside of the bracket clip depicted in FIG. **26**.
(32) FIG. **28** is a top plan view of another illustrative embodiment of a hinge pin support bracket and bracket clip for use in a hinge assembly as described herein.
(33) FIG. **29** is a side elevation view of the hinge pin support bracket and bracket clip depicted in FIG. **28** along with a portion of a base on which the hinge pin support bracket and bracket clip are mounted.
(34) FIG. **30** is a cross-sectional view of the hinge pin support bracket and bracket clip of FIGS. **28-29** taken along line **30-30** in FIG. **29**.
(35) FIG. **31** is a top plan view of another illustrative embodiment of a hinge pin support bracket and bracket clip for use in a hinge assembly as described herein.
(36) FIG. **32** is a side elevation view of the hinge pin support bracket and bracket clip depicted in FIG. **31** along with a portion of a base on which the hinge pin support bracket and bracket clip are mounted.
(37) FIG. **33** is a cross-sectional view of the hinge pin support bracket and bracket clip of FIGS.

31-32 taken along line **33-33** in FIG. **32**.

DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(38) In the following description of illustrative embodiments, reference is made to the accompanying figures of the drawing which form a part hereof, and in which are shown, by way of illustration, specific embodiments. It is to be understood that other embodiments may be utilized and structural changes may be made without departing from the scope of the present invention.

(39) The hinge assemblies described herein may be used with windows having one or more rotating sashes in a window frame, with each rotating sash being supported by at least one hinge assembly to control rotation of the sash relative to the frame. Some examples of windows with which the hinge assemblies described herein may be used include casement windows, awning windows, French casement windows, hopper windows, tilt-turn windows, pivot windows, utility windows, etc.

(40) One illustrative embodiment of a portion of a casement window is depicted in the perspective view of FIG. **1**. The casement window **2** includes a rotating sash **4** mounted in a frame including a frame member **8**. The rotating sash **4** is supported at its bottom end by a hinge assembly **10**, with the hinge assembly **10** being attached to a frame member **8** of the window frame in which the rotating sash **4** is located. Although not depicted, it will be understood that the sash **4** is also typically supported at its top end by a second hinge assembly.

(41) The hinge assembly **10**, which is isolated or removed from the sash **4** and window frame member **8** in FIG. **2**A, includes a track assembly **20**, sash arm **50**, and connecting arm **60**. The hinge assembly **10** is, in many respects, similar to typical hinge assemblies used to support rotating sashes in casement windows. For example, the track assembly **20** includes a base **22** and a shoe track **26**, with shoe **28** being slidably engaged with the shoe track **26** as known in conventional casement window hinge assemblies.

(42) The base **22** of track assembly **20** includes a shoe track end **24** and a hinge pin end **25**, with the base **22** extending between the shoe track end **24** and the hinge pin end **25** along track axis **21**. The shoe track **26** is also aligned with and extends along the track axis **21**, with the shoe track **26** extending from the shoe track end **24** of the base **22** towards the hinge pin end **25** of the base **22**. Typically, the shoe track **26** will not extend along the base **22** to the hinge pin end **25** and may, therefore, be described as occupying only a portion of the base **22** of the track assembly **20**.

(43) Sash arm **50** is, in the depicted illustrative embodiment, attached to the bottom **6** of the sash **4** and pivotally connected to the shoe **28** to both support and control movement of the sash relative to the window frame (e.g., window frame member **8**) during opening and closing of the sash **4**. Connecting arm **60** is pivotally connected to both the sash arm **50** and a hinge pin **30** of the track assembly **20** to further control movement of the sash arm **50** and, therefore, sash **4** relative to the window frame (e.g., window frame member **8**) as the sash **4** is opened and closed.

(44) In one or more embodiments, the connecting arm **60** may be described as extending from a hinge pin end pivotally connected to the hinge pin **30** and a sash end pivotally connected to the sash arm **50**. The sash end of the connecting arm **60** is pivotally connected to the sash arm **50** at an intermediate location spaced from the shoe end of the sash arm **50**.

(45) The pivotal connection between sash arm **50** and the shoe **28** along with the pivotal connections of the connecting arm **60** to both the hinge pin **30** and the sash arm **50** allow sash **4** to rotate as the sash is opened and closed, with the shoe **28** sliding along shoe track **26** during opening and closing of the sash **4**. As also known in conventional hinge assemblies used in casement windows, the shoe **28** engages the shoe track **26** such that the shoe **28** is restrained from moving transverse to the track axis **21** defined by the base **22** of the track assembly **20** as the shoe **28** slides along shoe track **26**.

(46) The hinge assembly **10** as depicted in FIG. **2**A is in an open configuration in which the sash connected to the sash arm **50** is in an open position relative to a window frame. FIG. **2**B depicts the components of the hinge assembly **10** of FIG. **2**A as arranged when a sash connected to the hinge

assembly **10** is in a closed configuration relative to a window frame. As depicted in FIG. **2**B, the sash arm **50** and connecting arm are rotated relative to their positions in FIG. **2**A such that the sash arm **50** and connecting arm **60** are generally aligned with the track axis **21** when the hinge assembly is in the closed configuration.

(47) As noted above, the connecting arm **60** is pivotally connected to both the sash arm **50** and the hinge pin **30**. As in conventional hinge assemblies used with, for example, casement windows, the track assembly **20** includes a hinge pin **30** attached to the base **22** of the track assembly **20**, with the connecting arm **60** being pivotally connected to the hinge pin **30** for rotation about the hinge pin **30**. With reference to FIG. **2**C, in one or more (but not all) embodiments, the hinge pin **30** includes a stud or base portion extending through the base **22** of the track assembly to a base end **32**, with the reminder of the hinge pin **30** extending away from the base **22** of the track assembly **20** to the upper end **34** of the hinge pin **30** located distal from the base **22**.

(48) Unlike conventional hinge assemblies, the illustrative embodiment of end assembly **10** depicted in FIG. **2**A includes a hinge pin support that, in the depicted illustrative embodiment, is in the form of a separate and discrete hinge pin support bracket **40**. The hinge pin support bracket **40** extends above the base **22** of the track assembly to support and restrain the upper end **34** (see, e.g., FIG. **2**C) of the hinge pin **30** from movement relative to the base **22** of the track assembly **20**. The hinge pin support bracket **40** is only one example of the structures that may be used to restrain the upper end **34** of the hinge pin **30** from movement relative to the base **22** of a track assembly **20**.

(49) Also, with reference to FIG. **2**C, the hinge pin end of the connecting arm **60** is located between the hinge pin support bracket **40** and the base **22** of the track assembly **20**. As a result, forces exerted on the hinge pin **30** by connecting arm **60** are distributed over a base portion of hinge pin **30** proximate the base end **32** and an upper portion of hinge pin **30** proximate the upper end **34**, rather than being concentrated over the base portion of the hinge pin **30** as would happen where the upper portion of the hinge pin **30** proximate the upper end **34** was not restrained by the hinge pin support bracket **40**.

(50) The illustrative embodiment of hinge pin support bracket **40** is depicted in isolation in FIGS. **3-4**. FIG. **3**, a top view of the hinge pin support bracket **40** depicts the hinge pin bore **42** in which the upper portion of the hinge pin **30** (i.e., the portion of the hinge pin **30** proximate the upper end **34**). The hinge pin support bracket **40** also includes a fastener opening **44** through which one or more mechanical fasteners may pass to attach the hinge pin support bracket **40** to the base **22** of the track assembly **20** and/or the window frame.

(51) Positioning of the hinge pin **30** in the hinge pin bore **42** provides the restraint of the hinge pin **30** relative to the base **22** because the hinge pin support bracket **40** is connected to the base **22** of the track assembly **20** (although, in some alternative embodiments, the hinge pin support bracket **40** may be attached directly to the frame member of a window frame to which the track assembly **20** is, itself, attached).

(52) In one or more embodiments, the hinge pin support bracket may be described as having a base portion **45** and a support portion **46**, with the base portion **45** being used to attach the hinge pin support bracket **40** to the base **22** of a track assembly and/or a window frame and the support portion **46** including the hinge pin bore **42** to restrain the upper portion of a hinge pin **30** located therein. In one or more embodiments, the support portion **46** may be described as being cantilevered over the hinge pin **30** supported by the hinge pin support bracket **40** to provide a gap between the support portion **46** and the base **22**, with the connecting arm **60** being positioned about the hinge pin **30** in the gap provided between the hinge pin support bracket **40** and the base **22** of the track assembly **20**.

(53) Another optional feature that may be provided in one or more embodiments of a hinge pin support bracket used in a hinge assembly as described herein can be discussed with reference to FIGS. **5**A-**5**B and **6**A-**6**B in which a portion of the base **22** proximate the hinge pin end **25** of the base **22** is depicted.

(54) As seen in FIGS. **5**A-**5**B, the support bracket **40** may be described as being aligned with the track axis **21** defined along base **22**, with the hinge pin **30** extending from the base **22** into the hinge pin bore **42** in hinge pin support bracket **40**. When so positioned, the bracket fastener opening **44** provided in the hinge pin support bracket **40** is aligned with a fastener opening **23** in the base **22** of the track assembly **20**. Inserting a fastener through bracket fastener opening **44** and fastener opening **23** in base **22** would secure the hinge pin support bracket **40** and allow the hinge pin support bracket **42** restrain the upper portion of the hinge pin **30** as described herein.
(55) In many embodiments of hinge assemblies used with, for example, casement windows, the hinge pin **30** may be eccentrically shaped or mounted such that rotation of the hinge pin about its hinge pin axis (see, for example, axis **31** of hinge pin **30** in FIG. **2**C) moves the position of the hinge pin relative to the base of a track assembly to allow for field adjustment of the hinge assembly as needed. In the depicted embodiment, hinge pin **30** includes wrench flats **36** that may be used to rotate the hinge pin **30** as needed.
(56) The eccentric design of one or more (but not all) embodiments of hinge pins used in hinge assemblies as described herein may be seen, for example, in the cross-sectional view the illustrative embodiment of hinge pin **30** of FIGS. **5**B and **6**B. In the depicted illustrative embodiment, the stud or base portion of the hinge pin **30** extends through base **22** of the track assembly **20** to the base end **32** and defines a stud axis **31** around which the hinge pin **30** can be rotated if desired.
(57) The rotational position of the stud or base portion of the hinge pin **30** relative to hinge pin axis **31** may, in one or more embodiments, be retained by friction between the stud or base portion of the hinge pin **30** and the base **22** of the track assembly. In one or more embodiments, the portion of the base **22** of the track assembly surrounding the opening through which the hinge pin **30** extends may be textured or otherwise treated to increase frictional engagement between the base **22** and the hinge pin **32** reduce the likelihood of unwanted rotation of the hinge pin **30** relative to the base **22**. For example, if the hinge pin is deformed to form a swedge (as is common in conventional hinge pins), the portion of the base surrounding the opening through which the hinge pin extends and which would be in contact with the swedge may be textured, roughened, etc. to increase friction between the swedge of the hinge pin and the base **22**. Increasing friction between the hinge pin and the base of the track assembly may be useful to, for example, limit unwanted pin rotation (sometimes referred to as pin migration) if the sash is subjected to buffeting due to wind or other repetitive forces. Other techniques of fixing the rotational position of the hinge pins used in hinge assemblies as described herein may be used in place of, or in addition to, friction.
(58) The pin portion of the hinge pin **30**, i.e., the portion of the hinge pin **30** located above the base **22** of the track assembly **20** defines a pin axis **33** that is offset from the stud axis **31**, i.e., the stud axis **31** and the pin axis **33** are generally aligned with each other, but are offset from each other in a plane defined by the track axis **21** and one or both of the stud axis **31** and the pin axis **33** (e.g., the top of the base **22** in the depicted illustrative embodiment).
(59) The connecting arm **60** is attached to the pin portion of the hinge pin **30** and pivots/rotates about the pin axis **33**. Because the stud axis **31** and the pin axis **33** are offset from each other, rotation of the hinge pin **30** about stud axis **31** (i.e., rotation of the stud portion relative to the base **22**) moves the pin portion (including the upper end **34** of the hinge pin **30**) and its pin axis **33** towards or away from, for example, the hinge pin end **25** of the base **22**. Because the connecting arm **60** is pivotally/rotatably attached to the pin portion for rotation about the pin axis **33**, movement of the pin portion and its pin axis **33** also moves the hinge pin end of the connecting arm **60** to adjust the position of a sash connected to the hinge assembly.
(60) Adjusting the position of the hinge pin **30** relative to the track axis **21** may, however, require some adjustment in the position of the hinge support bracket **40**. With reference to FIGS. **6**A-**6**B, the hinge pin support bracket **40** includes an elongated noncircular opening as its bracket fastener opening **44** to allow for movement and/or rotation of the hinge pin support bracket **40** relative to the track axis **21** to accommodate adjustment of the position of the hinge pin **30** relative to the track

axis **21**. The resulting position of the hinge pin support bracket **40** may result in a hinge pin support bracket **40** that is canted or somewhat misaligned with the track axis **21**, but which still provides restraint to the hinge pin **30** as described herein.

(61) FIG. **7-9** depict an alternative illustrative embodiment of a hinge pin support bracket **140** that may be used in one or more embodiments of a hinge assembly as described herein (with FIG. **8** being an enlarged cross-sectional view of the hinge pin support block of FIG. **7** taken along line **8-8**, with a portion of a base **122** to which the hinge pin support block **140** would be attached). The hinge pin support bracket **140** includes a base portion **145** and a support portion **146**. The support portion **146** includes a hinge pin bore **142** formed therein. The base portion **145** includes a fastener opening **144** configured to receive a fastener to secure the hinge pin support bracket **142** the base of a track assembly (or directly to a window frame to which the track assembly itself is attached).

(62) With reference to the enlarged cross-sectional view of FIG. **8**, another optional feature of one or more embodiments of a hinge pin support bracket that may be used in one or more embodiments of a hinge assembly as described herein is found in the interlocking mechanical features provided in both the base **122** of the track assembly and the hinge pin support bracket **140**.

(63) In the depicted illustrative embodiment, the hinge pin support bracket **140** includes a flange **147** sized and shaped to fit within a corresponding fastener opening **123** in the base **122**. As a result, the flange **147** and the opening **123** provide one illustrative embodiment of interlocking mechanical features that may enhance transfer of forces from the hinge pin support block **142** the base **122** of a track assembly on which the hinge pin support bracket **140** is mounted.

(64) Flange **147** on hinge pin support block **140** and opening **123** in base **122** are only one example of the myriad of interlocking mechanical features that may be provided between the hinge pin support bracket **140** and the base **122**, some examples of which may include, but are not limited to, any complementary protrusions and recesses formed in the facing surfaces of hinge pin support block **140** and the base **122** (e.g., one or more sets of complementary pins and apertures), etc.

(65) Although interlocking mechanical features may be used to transfer loads between the hinge pin support block and a base to which the block is attached, other techniques and/or features may be used to enhance that load transfer in the absence of interlocking mechanical features. For example, adhesives, structured surfaces (e.g., knurled surfaces, etc.), abrasive materials, etc. may be used to enhance friction between the hinge pin support block and a base to which the block is attached, with that friction providing an improved transfer of loads from the support block to the base.

(66) Another feature that may be found in track assemblies including a hinge pin support block as described herein can be found in the shoulder **149** provided on hinge pin support block **140** that, in the depicted embodiment, is adapted to nest with the hinge pin end **125** of the base **122**. The shoulder **149** may, in one or more embodiments, be useful in accurately positioning the hinge pin support block **140** relative to the base **122** of a track assembly to which the hinge pin support block **140** is attached.

(67) FIGS. **10-11** depict another illustrative embodiment of a hinge pin support block that may be used in one or more embodiments of a hinge assembly as described herein. The hinge pin support block **240** is depicted alone in FIG. **10** and, in FIG. **11**, positioned on a base **222** of a track assembly that may be used in one or more embodiments of a hinge assembly as described herein.

(68) The hinge pin support block **240** extends from a first end **241** to a second end **243** and includes one or more openings **244** to facilitate the use of mechanical fasteners to attach the hinge pin support block **242** the base **222** of a track assembly **220** of a hinge assembly as described herein. With reference to FIG. **11**, the first end **241** of the hinge pin support block **240** may also serve as a mechanical stop for a shoe that is slidably engaged with the shoe track **226** provided in the track assembly **220**.

(69) Hinge pin support block **240** also includes a support portion **246**, with a hinge pin bore **242** formed in the support portion **246**. In some respects, the remainder of the hinge pin support block **240** may be described as the base portion of the hinge pin support block **240** because it is used to

position the support portion **246** over a hinge pin **230** attached to the base **222** of the track assembly **220**. The hinge pin support block **240** may further include an access aperture **248** through which the hinge pin **230** extends when the hinge pin is attached to the base **222** and positioned such that its upper portion is restrained within the hinge pin board **242** in the support portion **246**.
(70) Another alternative illustrative embodiment of a hinge pin support is depicted in connection with FIGS. **12-14**, with only a portion of the track assembly **320** being depicted. Among the features depicted are a portion of the base **322** extending away from the hinge end **325** and a portion of the shoe track **326** of the track assembly **320**.
(71) The portion of track assembly **320** depicted in these figures includes a hinge pin support in the form of a support portion **346** positioned above the base **322** of the track assembly **320**, with the support portion **346** including a hinge pin bore **342** used to restrain the upper portion of a hinge pin **330** extending upwards away from the base **322** in a manner similar to that described in connection with other illustrative embodiments of track assemblies as described herein.
(72) Unlike hinge pin support blocks that are provided as separate and discrete articles attached to or positioned relative to the base **322** of the various track assemblies described herein, the support portion **346** is formed as an integral portion of the base **322** which is formed, bent, molded, shaped, etc. such that the support portion **346** is located above a portion of the base **322** to provide support to the upper portion of a hinge pin **330** (i.e., the portion of the hinge pin **330** proximate the upper end **334** of the hinge pin **330**) with its base end **332** attached to the base **322** of the track assembly **320**.
(73) In one or more embodiments, the support portion **346** may be described as a hinge pin support tab formed integrally with the base **322**, with the hinge pin support tab **346** and the base **322** defining a U-shaped structure that includes a bottom from which to legs extend, with the hinge pin **330** extending between the legs of the U-shaped structure distal from the bottom of the U-shaped structure. With reference to, e.g., FIGS. **13-14**, the bottom of such a U-shaped structure may correspond to the hinge pin end **325** with one of the legs being formed by the base **322** and the other leg being formed by the hinge pin support tab **346**, both of which extend away from the injection and **325** of the track assembly **320**.
(74) To provide further structural integrity to the support portion **346**, one or more embodiments of such a design may include a bushing **370** and mechanical fastener **372** extending through a fastener opening **344** in the support portion **346**. The mechanical fastener **372** may be secured to the remainder of the base **322** of track assembly **320** and/or any underlying structure such as, e.g., a window frame, etc.
(75) FIGS. **15-18** depict another illustrative embodiment of a hinge pin support block **440** that may be used in connection with one or more embodiments of a hinge assembly as described herein. With reference to FIGS. **15-16** (where FIG. **16** is a cross-sectional view of FIG. **15** taken along line **16-16**), the hinge pin support block **440** includes a base portion **445** and a support portion **446** similar to many of the other illustrative embodiments of hinge pin support blocks described herein.
(76) A fastener opening **444** is provided in the base portion **445** of hinge pin support block **440**. Base portion **445** also includes a flange **447** which may form a portion of an interlocking mechanical structure as described in connection with hinge pin support block **140**. The fastener opening **444** can be used to secure the hinge pin support block **442** the base of a track assembly and/or to a window frame to which the base of the track assembly is attached.
(77) Hinge pin support block **440** is, in the depicted illustrative embodiment, includes a core **490** positioned in a body **492**. In one or more embodiments, body **492** may be made of one or more different polymeric materials, while core **490** may be formed of one or more different metals or other generally more mechanically rigid materials. This two-piece construction is, however, optional and may not be used in all embodiments.
(78) The hinge pin support block **440** may include a hinge pin bore **442** used to restrain a hinge pin used in connection with the hinge pin support block **440**. Unlike other embodiments of hinge pin

support blocks described herein, however, the hinge pin support block **440** includes a bushing **480** in the hinge pin bore **442**. Bushing **480** includes a bushing aperture **482** with the pin portion of an eccentric hinge pin used in connection with the hinge pin support block **440** being positioned in the bushing aperture **482** of bushing **480**.

(79) The bushing aperture **482** and the pin portion of a hinge pin located in the bushing aperture **482** are positioned off-center in the hinge pin bore **442** of hinge pin support block **440**. As a result, rotation of an eccentric hinge pin (e.g., a hinge pin similar to hinge pin **30** depicted in FIGS. **5**B & **6**B) about a stud axis **431** (extending through the center of the hinge pin bore **442**) rotates the bushing **480** about the stud axis **431** in the hinge pin bore **442** and moves the pin portion of a hinge pin (and its associated pin axis **433**) located in the bushing aperture **482** towards or away from a shoe track to which the hinge pin and the hinge pin support block **440** are attached (in other words, moves the upper portion of the hinge pin along a track axis defined by the track assembly as described herein).

(80) Another feature depicted in connection with the illustrative embodiment of hinge pin support block **440** is the use of visible indicia indicative of the rotational position of the bushing **480** in the hinge pin bore **442** of the hinge pin support block **440**. In particular, the visible indicia may take the form of an indicator mark **496** on the bushing **480** and markings **498** on the hinge pin support block **440**. In one or more embodiments, the indicator mark **496** and markings **498** may be used together to indicate a position of the pin portion of an eccentric hinge pin relative to a track axis of a track assembly as described herein. As a result, a user may be more able to accurately adjust the position of the pin portion of the hinge pin (and its associated pin axis **433**) relative to the track axis using the visible indicia. In the home or zero position, the pin axis **433** may be aligned with the stud axis **431** when viewed from a side as seen in both FIGS. **15** and **17**. Rotation of the bushing **480** and the stud portion of the hinge pin about the stud axis **431** of the hinge pin moving the pin portion of the hinge pin (and its pin axis **433**) towards or away from, for example, the shoe track of a track assembly.

(81) Another optional feature that may be used in one or more embodiments of hinge pin support brackets of one or more embodiments of hinge assemblies as described herein is depicted in connection with FIGS. **19-20**.

(82) One illustrative embodiment of a hinge pin support bracket **540** is depicted in FIG. **19**. Hinge pin support bracket **540** includes, as described in connection with any other illustrative embodiments of hinge pin support brackets described herein, includes a hinge pin bore **542** and a fastener bore **544**. As described in connection with other illustrative embodiments of hinge pin support brackets, the hinge pin bore **542** is configured to restrain the upper portion of a hinge pin used in a hinge assembly, while the fastener bore **544** can be used to secure the hinge pin support bracket **542** the base of a track assembly of a hinge assembly as described herein.

(83) The hinge pin support bracket **540** also includes a vent limiter **590** attached thereto which can be used to limit rotation of a connecting arm attached to a hinge pin restrained within the hinge pin support bore **542**. In particular, vent limiter **590** includes a stop surface **592** against which a connecting arm acts as a hinge assembly is moved from a closed configuration to an open configuration.

(84) More specifically, with reference to FIG. **20**, the hinge pin support bracket **540** including the vent limiter **590** are depicted as attached to the base **522** of a track assembly **520** of a hinge assembly as described herein. The track assembly includes a base **522** and shoe track **526** both of which extend along a track axis **521** in a manner similar to other track assemblies as described herein. Also depicted in FIG. **20** are a hinge pin **530** restrained within the hinge pin bore **542** and a connecting arm **560** pivotally connected to the hinge pin **530**. Connecting arm **560** extends along a connecting arm axis **560** away from the hinge pin **530** and towards a sash arm as described herein in connection with other illustrative embodiments of hinge assemblies.

(85) As depicted in FIG. **20**, connecting arm **560** is in an open position such that connecting arm

560 acts against the stop surface **592** of vent limiter **590**, thereby limiting further rotation of the connecting arm **560** about hinge pin **530**. Limiting that rotation will, in turn, limit the degree to which a sash connected to the track assembly **520** can be opened within a window frame. Limiting opening of a sash can be useful for safety and security reasons, as well as to limit the likelihood of damage to a window sash if the sash is opened past a selected position.
(86) In the depicted illustrative embodiment of FIGS. **19-20**, the vent limiter **590** and the hinge pin support bracket **540** are depicted as being separate and discrete articles that are attached to each other by one or more, in particular two, mechanical fasteners that may be received in bores **594** that extend through the vent limiter **590** and into the hinge pin support bracket **540**. Such mechanical fasteners may include, but are not limited to, screws, bolts, rivets, etc.
(87) Alternatively, the hinge pin support bracket **540** and vent limiter **590** may be attached to each other by any suitable technique or combination of techniques including, mechanical fasteners, adhesives, welding, mechanically interlocking connections, etc. in still other alternative embodiments, the hinge pin support bracket **540** and vent limiter **590** may be manufactured together as one integral hinge pin support bracket including a vent limiter, with the hinge pin support bracket including integral vent limiter being manufactured by any suitable technique or combination of techniques including, casting, forging, machining, etc.
(88) Yet another optional feature that may be provided in one or more embodiments of the hinge assemblies described herein is depicted in connection with FIG. **21**. In particular, the illustrative embodiment depicted in FIG. **21** includes a base **622** of a track assembly with a hinge pin **630** attached to the base **622** as described herein. The base **622** of the track assembly extends along a track axis **621**, with only a portion of the base **622** including a hinge pin end **625** and extending towards the shoe end of the base **622** being depicted in FIG. **21**.
(89) A portion of a connecting arm **660** used in the hinge assembly is depicted as being attached to the hinge pin **630** for rotation about the pin axis **633** in much the same manner as described in connection with rotation of connecting arms of hinge assemblies around pin axes defined by hinge pins as described herein.
(90) The hinge assembly is also depicted as including a hinge pin support in the form of a hinge pin support bracket **640**. The depicted illustrative embodiment of hinge pin support bracket **640** includes a base portion that may be attached to the base **622** as described in connection with other illustrative embodiments of hinge pin support brackets described herein. The hinge pin support bracket **640** also includes a support portion that extends above the base **622** and includes, as described in connection with other illustrative embodiments of hinge pin support brackets described herein, a hinge pin bore through which the hinge pin **630** extends.
(91) One difference between the hinge pin **630** and hinge pin support bracket **640** depicted in FIG. **21** as compared to other illustrative embodiments of hinge pin supports and hinge pins as described herein is that the hinge pin **630** protrudes above the hinge pin support bracket. The hinge pin **630** includes a ring slot **638** position above the support portion of the hinge pin support bracket such that the hinge pin support bracket **640** can be placed over the hinge pin **632** restrain the hinge pin **630** while still allowing a user to install a C-clip or other retaining feature on the hinge pin **630** as needed. As a result, the hinge pin **630** may be described as having a protruding portion **639** located above the support portion of the hinge pin support bracket **640**. This design may, in one or more embodiments, simplify installation of the hinge assembly using an extended hinge pin in combination with a hinge pin support as described herein.
(92) A portion of another illustrative embodiment of a hinge assembly for supporting a rotating sash in a window frame as described herein is depicted in FIGS. **22-27**. In particular, FIG. **22** is a perspective view of the components of the hinge assembly used to provide support for a hinge pin **730** in their assembled state. Although not depicted in FIGS. **22-27**, it should be understood that the other components of hinge pin assemblies as described herein and depicted in the figures would be provided in a hinge assembly incorporating the components depicted in FIGS. **22-27** and described

in connection with those figures.
(93) The depicted hinge assembly includes a base **722** located on a window frame member **708**, with the base **722** extending along a track axis **721** as described in connection with other bases and track axes described herein. Hinge pin **730** is attached to the base **722** as described herein in connection with other illustrative embodiments. Also depicted in FIG. **22** is a portion of a connecting arm **760** rotatably attached to the hinge pin **730**, with the connecting arm **760** rotating about hinge pin axis **733** defined by the hinge pin **730**.
(94) The upper end of the hinge pin **730** is supported by the depicted embodiment of hinge pin support bracket **740** which includes a hinge pin bore **742** in which the upper portion of the hinge pin **730** is located. The hinge pin support bracket **740** is attached to the base **722** and the window frame member **708** using, in the depicted illustrative embodiment, a threaded fastener **772** (although any other suitable fastener could be used in place of a threaded fastener **772** as described herein).
(95) In the illustrative embodiment depicted in FIGS. **22-27**, a bracket clip **790** is provided in addition to the hinge pin support bracket **740**. In one or more embodiments of hinge assemblies as described herein that include a bracket clip such as bracket clip **790**, the bracket clip **790** may extend past the hinge pin end **725** of the base **722** as seen in, e.g., FIG. **22**.
(96) FIG. **23** is an exploded perspective view of the hinge pin support bracket **740**, bracket clip **790**, and fastener **772** of the hinge pin assembly depicted in FIG. **22**. As discussed above, the hinge pin bracket **740** includes a hinge pin bore **742** configured to receive the upper end of a hinge pin as described herein. Hinge pin support bracket **740** also includes a fastener opening **744**. The fastener **772** defines an axis **771** extending therethrough, with the bracket clip **790** and hinge support bracket **740** being, in one or more embodiments, configured to rotate about the axis **771** as described herein.
(97) In the depicted illustrative embodiment of hinge pin bracket **740**, the bracket includes a base portion **745** and a support portion **746** as described in connection with, for example, hinge pin support bracket **40** depicted in FIGS. **3-4**. The base portion **745** of hinge pin support bracket **740** is used to attach the hinge pin support bracket **740** to the base **722** and/or the window frame **708**. The support portion **746** includes a hinge pin bore **742** two restrain the upper portion of a hinge pin (e.g., hinge pin **730**) located therein. The support portion **746** may be described as being cantilevered over the hinge pin that is supported by the support portion of the hinge pin support bracket **742** provide a gap between the support portion **746** and the base **722**, with the connecting arm **760** being positioned about the hinge pin **730** in the gap provided between the hinge pin support bracket **740** and the base **722**.
(98) The bracket clip **790** is also depicted in FIG. **23**. The depicted embodiment of bracket clip **790** includes a base portion **792** from which a boss **794** protrudes. In particular, the boss **794** protrudes from the underside of the body portion **792** of the bracket clip **790** (where the underside of the body portion **792** is the surface of the bracket clip **790** facing the base portion **745** of the hinge support bracket **740**).
(99) Boss **794** is, in the depicted embodiment, configured to fit within the bracket opening **744** in the base portion **745** of the hinge support bracket **740**. The boss **794** protruding from the base portion **792** of bracket clip **790** and corresponding bracket opening **744** in base portion **745** of hinge support bracket **740** depict only one embodiment of interlocking mechanical features found in the base portion **745** of the hinge support bracket **740** and the bracket clip **790** that cooperate to prevent rotation of the hinge pin support bracket **740** and the bracket clip **790** relative to each other about an axis extending through the base **722**, the base portion **745** of the hinge pin support bracket **740** and the bracket clip **790** (see, e.g., FIG. **22**). In the depicted embodiment, that axis may be defined by the axis **771** extending through the fastener **772**.
(100) The complementary elongated shapes of both the bracket opening **744** in base portion **745** of hinge pin support bracket **740** and the boss **794** protruding from the base portion **792** of bracket

clip **790** illustrate only one set of complementary shapes that form interlocking mechanical features between the hinge pin support bracket and bracket clips of hinge assemblies as described herein. Many other alternative interlocking mechanical shapes configured to prevent rotation between the hinge pin support bracket **740** and the bracket clip **790** relative to each other are, of course, possible.

(101) In the depicted illustrative embodiment of bracket clip **790**, a flange **797** protrudes from the boss **794** of the bracket clip **790**. Flange **797** is, in one or more embodiments, configured to fit within a fastener opening provided in the base **722** of the hinge assembly including bracket clip **790**. The relationship between flange **797** and a fastener opening **723** in base **722** is seen in the partial cross-sectional view of FIG. **25**. In a manner similar to that discussed in connection with the hinge pin support bracket **140** which includes a flange **147** sized and shaped to fit within a corresponding fastener opening **123** in base **122** as seen in FIG. **8**, flange **797** on bracket clip **790** is sized and shaped to fit within a corresponding fastener opening **723** in base **722** as depicted in FIG. **25**.

(102) The flange **797** and opening **723** provide only one illustrative embodiment of interlocking mechanical features that may be provided to transfer loads between the hinge pin support bracket **740** and the base **722** (as well as the underlying frame member **708**). Those loads may be generated on the hinge pin **730** through connecting arm **760** as described elsewhere herein.

(103) Taken together, the interlocking mechanical features used to transfer loads to the base of a hinge assembly (and/or the frame member on which that hinge assembly is mounted) from a hinge pin using a hinge pin support bracket can be, in one illustrative embodiment, found in the combination of the flange **797** and opening **723** in base **722** as well as the boss **794** on bracket clip **790** and the opening **744** in the base portion **745** of the hinge pin support bracket **740**. Both sets of complementary interlocking mechanical features found in that set of components provides a load transfer path from the top of the hinge pin to the base **722** and/or the frame member on which the hinge assembly is mounted.

(104) Further, it should be noted that the complementary shapes of the flange **797** and opening **723** in base **722** (i.e. the interlocking mechanical features) allow for rotation of the bracket clip **790** and hinge pin support bracket **740** about the axis **771** passing through, in the depicted embodiment, the fastener **772**. That rotation may be useful where, for example, the position of the hinge pin **730** (see, e.g., FIG. **22**) relative to the base **722** needs to be adjusted for proper closure of a sash supported by the depicted hinge assembly as described herein. Those types of adjustments are, for example, described in connection with the hinge pin support bracket **40** and hinge pin **30** depicted in FIGS. **5**A-**6**B. Adjusting the position of an eccentrically mounted hinge pin **730** on base **722** can often only be accomplished by rotation of the hinge pin support bracket **740** and bracket clip **790** about axis **771** extending through fastener **772** (in addition to rotating the eccentrically mounted hinge pin).

(105) In addition to providing for rotation of the hinge pin support bracket **740** about an axis passing through the base **722**, base portion **745** of hinge pin support bracket **740** and bracket clip **790** (such as, for example, axis **771** passing through fastener **772** in the depicted embodiment) to allow for adjustment of the location of a hinge pin **730** on base **722**, it may be desirable to limit the amount of any such rotation.

(106) In the depicted illustrative embodiment, bracket clip **790** may include features that are configured to allow for some degree of rotation of the bracket clip and, therefore, the hinge pin support bracket relative to an axis passing through the base **722**, base portion **745**, and bracket clip **790** (such as, for example, axis **771**). In particular, bracket clip **790** includes, as described above, a base portion **792**, but also includes a mechanical stop **794** that is configured to act on the hinge pin end **725** of the base **722** when the bracket clip **790** and hinge pin support bracket **740** are assembled on the base **722** (as seen in, for example, FIG. **22**).

(107) In one or more embodiments, the mechanical stop **794** of bracket clip **790** may include one or

more surfaces **795** and **795′** configured to act on the hinge pin end **725** of the base **722**. In the depicted illustrative embodiment, central surface **795′** may be oriented perpendicular to the track axis **721** defined along the length of the base **722** while the corresponding hinge pin end **725** of base **722** is also oriented perpendicular to the track axis **721**. The depicted illustrative embodiment of bracket clip **790** also includes, however, side surfaces **795** that are not oriented perpendicular to the track axis **721**. As a result, rotation of the bracket clip **790** about axis **771** (and corresponding rotation of the hinge pin support bracket **740**) is allowed up to a selected limit. With reference to FIG. **27**, the side surfaces **795** may allow for rotation of the bracket clip **790** (and corresponding rotation of the hinge pin support bracket **740**) over an angle α (alpha) of, for example, 10° or less, 8° or less, 6° or less, 4° or less, or 2° or less.

(108) Although the depicted embodiment of bracket clip **790** includes side surfaces **795** oriented at the same angle relative to the track axis **721**, one or more alternative embodiments of hinge assemblies as described herein may include a bracket clip **790** that include side surfaces provided with different angular orientations relative to the track axis. Moreover, although the depicted embodiment of bracket clip **790** includes a pair of side surfaces, one or more alternative embodiments of bracket clips as described herein may include only one side surface oriented at an angle relative to the track axis that allows for rotation of the bracket clip and hinge pin support bracket as described herein. Further, although the depicted embodiment of bracket clip **790** includes a central surface **795′** that is aligned with the edge at the hinge pin end **725** of the base **722**, one or more alternative embodiments of bracket clips as described herein may not include an aligned surface such as central surface **795′**.

(109) In still other variations, the hinge pin end **725** of the base **722** may not be perpendicular to the track axis **721** but may, instead, include features configured to allow for rotation of the bracket clip **790** and hinge pin support bracket **740** as needed to adjust the position of a hinge pin **730**. In such embodiments, the mechanical stop **794** provided on bracket clip **790** may have any shape complementary to the shape of the hinge pin end **725** a base **722** needed to control rotation of the bracket clip **790** and hinge pin support bracket **740** relative to an axis extending through those components as well as the base of the hinge assembly. For example, the hinge pin end **725** of base **722** may include one or more surfaces oriented at a non-perpendicular angle relative to the track axis **721** while the mechanical stop **794** includes, for example, a single flat surface configured to interact with the hinge pin end **725** of base **722** in a manner that allows for rotation of the bracket clip **790** and hinge pin support bracket **740** as described herein.

(110) While the boss **794** on bracket clip **790** and the opening **744** in the hinge pin support bracket **740** along with flange **797** on bracket clip **790** and a corresponding opening **723** in base **722** provide one illustrative embodiment of a set of interlocking mechanical features used to control rotation between the bracket clip **790** and hinge pin support bracket **740** as well as provide for load transfer from a hinge pin through the hinge pin support bracket **742** the base **722** and/or an underlying window frame member, many other alternative interlocking mechanical features may perform those same functions.

(111) For example, FIGS. **28-30** depict another illustrative embodiment of a hinge pin support bracket and bracket clip for use in a hinge assembly as described herein that includes a set of interlocking mechanical features configured to control rotation between a bracket clip and hinge pin support bracket, as well as to provide for load transfer from a hinge pin supported by the hinge pin support bracket to a base and/or underline window frame member on which the hinge assembly including these components is mounted.

(112) In particular, a hinge pin support bracket **840** and bracket clip **890** are provided to support the upper portion of a hinge pin as described in connection with other illustrative embodiments of hinge pin support brackets described herein. The hinge pin support bracket **840** includes a hinge pin opening **842** and extends along a track axis **821**. The bracket clip **890** of FIGS. **28-30** includes a fastener opening **893** through which a fastener opening axis **871** extends, with axis **871** being

comparable to axis **771** depicted in connection with the illustrative embodiment of hinge pin support bracket **740** and bracket clip **790** depicted in FIGS. **22-27**.
(113) With reference to FIG. **29**, the hinge pin support bracket **840** is mounted on a base **822** which includes a hinge pin end **825**. Bracket clip **890** includes a mechanical stop **896** that interacts with hinge pin end **825** of base **822** in the same manner as discussed above in connection with mechanical stop **796** on bracket clip **790** and hinge pin end **725** of base **722**. Moreover, the variations discussed with respect to mechanical stop **796** and hinge pin end **725** would apply equally in this alternative embodiment of mechanical stop **896** and hinge pin end **825**.
(114) Control over relative rotation between the bracket clip **890** and the hinge pin support bracket **840** is provided in the embodiment depicted in FIGS. **28-30** by interlocking mechanical features in the form of the body **892** of bracket clip **890** and a pair of flanges **849** provided on hinge pin support bracket **840**. In particular, the body **892** of bracket clip **890** nests within the flanges **849** of hinge pin support bracket **840** such that rotation of the hinge pin support bracket **840** relative to the bracket clip **890** about axis **871** is prevented. Although the depicted embodiment includes a pair of flanges **849**, in one or more alternative embodiments, only one flange **849** may be provided in may be sufficient to provide control over relative rotation between the hinge pin support bracket **840** and bracket clip **890**.
(115) In the embodiment of bracket clip **890**, load transfer from the hinge pin support bracket **840** to the base **822** (and/or an underlying frame member on which these components are mounted) may be accomplished through a flange **897** extending into opening **823** in base **822** as seen in, e.g., FIG. **30**. As discussed above, the flange **897** and opening **823** in which the flange **897** is received may be circular to allow for rotation of the bracket clip **890** and hinge pin support bracket **840** about axis **871** to allow for adjustment of the position of a hinge pin supported by the hinge pin support bracket **840** as discussed herein in connection with other embodiments.
(116) Another illustrative embodiment of a hinge pin support bracket and bracket clip that may be used in a hinge assembly as described herein is depicted in FIGS. **31-33**. The hinge pin support bracket and bracket clip depicted in those figures includes another alternative set of interlocking mechanical features configured to control rotation between a bracket clip and hinge pin support bracket, as well as to provide for load transfer from a hinge pin supported by the hinge pin support bracket to a base and/or underline window frame member on which the hinge assembly including these components is mounted.
(117) In particular, a hinge pin support bracket **940** and bracket clip **990** are provided to support the upper portion of a hinge pin as described in connection with other illustrative embodiments of hinge pin support brackets described herein. The hinge pin support bracket **940** includes a hinge pin opening **942** and extends along a track axis **921**. The bracket clip **990** of FIGS. **31-33** includes a fastener opening **993** through which a fastener opening axis **971** extends, with axis **971** being comparable to axis **771** depicted in connection with the illustrative embodiment of hinge pin support bracket **740** and bracket clip **790** depicted in FIGS. **22-27**.
(118) With reference to FIG. **32**, the inch pin support bracket **940** is mounted on a base **922** which includes a hinge pin end **925**. Bracket clip **990** includes a mechanical stop **996** that interacts with the hinge pin end **925** of base **922** in the same manner as described above in connection with mechanical stop **796** on bracket clip **790** and hinge pin end **725** of base **722**. Moreover, the variations discussed with respect to mechanical stop **796** and hinge pin end **725** would apply equally in this alternative embodiment of mechanical stop **996** and hinge pin end **925**.
(119) Control over relative rotation between the bracket clip **990** and the hinge pin support bracket **940** is provided in the embodiment depicted in FIGS. **31-33** by interlocking mechanical features in the form of a flange **949** provided in the hinge pin support bracket **940** proximate the end of the hinge pin support bracket **940** that is near the hinge pin end **925** of the base **922**. The flange **949** may, for example, extend across a width of the hinge pin support bracket **940** in a direction generally perpendicular to the track axis **921**. The flange **949** has a shape complementary and is

received within a slot **999** provided in the bracket clip **990**. The flange **949** of hinge pin support bracket **940** and complementary slot **999** provided in the bracket clip **990** are seen in the top plan view of FIG. **31** (with the speakers being depicted in broken lines) as well as in the side elevation view of FIG. **32**. Although the depicted embodiment includes a flange **949** that extends across the entire width of the hinge pin support bracket **940**, in one or more alternative embodiments, the flange **949** may extend only over a portion of the width of the hinge pin support bracket **940**. Other variations will, of course, also be possible.

(120) In the embodiment of bracket clip **990**, load transfer from the hinge pin support bracket **940** to the base **922** (and/or an underlying frame member on which these components are mounted) may be accomplished through a flange **997** extending into opening **923** in base **922** as seen in, e.g., FIG. **33**. As discussed above, the flange **997** and opening **923** in which the flange **997** is received may be circular to allow for rotation of the bracket clip **990** and hinge pin support bracket **940** about axis **971** to allow for adjustment of the position of a hinge pin supported by the hinge pin support bracket **940** as discussed herein in connection with other embodiments.

(121) The complete disclosure of the patents, patent documents, and publications identified herein are incorporated by reference in their entirety as if each were individually incorporated. To the extent there is a conflict or discrepancy between this document and the disclosure in any such incorporated document, this document will control.

(122) Illustrative embodiments of hinge assemblies and methods are discussed herein with some possible variations described. These and other variations and modifications in the invention will be apparent to those skilled in the art without departing from the scope of the invention, and it should be understood that this invention is not limited to the illustrative embodiments set forth herein. Accordingly, the invention is to be limited only by the claims provided below and equivalents thereof. It should also be understood that this invention also may be suitably practiced in the absence of any element not specifically disclosed as necessary herein.

Claims

1. A hinge assembly for supporting a rotating sash in a window frame, the hinge assembly comprising: a track assembly configured for attachment to a window frame, the track assembly comprising a base, a shoe track, and a hinge pin, wherein the base extends along a track axis from a hinge pin end to a shoe track end, wherein the shoe track extends along the track axis, and wherein the shoe track extends from the shoe track end towards the hinge pin end, wherein the hinge pin comprises a base end connected to the base proximate the hinge pin end of the base and an upper end distal from the base; a shoe slidably engaged with the shoe track, wherein the shoe is configured to slide along the shoe track end the shoe is restrained from moving transverse to the track axis when the shoe is slidably engaged with the shoe track; a sash arm configured for attachment to a rotating sash, the sash arm extending from a shoe end to a distal end, wherein the shoe end is pivotally coupled to the shoe; a connecting arm extending from a hinge pin end to a sash end, wherein the sash end is pivotally connected to the sash arm at an intermediate location spaced from the shoe end of the sash arm, and wherein the hinge pin end of the connecting arm is pivotally connected to the hinge pin for rotation about a pin axis extending through the base end and the upper end of the hinge pin; and a hinge pin support attached to the base of the track assembly, the hinge pin support comprising a support portion extending above the base of the track assembly, wherein the support portion comprises a hinge pin bore, wherein a portion of the hinge pin is located in the hinge pin bore such that the hinge pin support restrains the upper end of the hinge pin from movement relative to the base, and wherein hinge pin extends through an opening in the connecting arm proximate the hinge pin end of connecting arm such that the hinge pin end of the connecting arm is located between the support portion and the base of the track assembly.
2. A hinge assembly according to claim 1, wherein the hinge pin support comprises a hinge pin

support bracket separate and discrete from the base of the track assembly.

3. A hinge assembly according to claim 2, wherein the hinge pin support bracket and the base of the track assembly comprise interlocking mechanical features configured to transfer mechanical loads directed along the track axis on the hinge pin to the base and the hinge support bracket.

4. A hinge assembly according to claim 2, wherein the base comprises a base fastener aperture and wherein the hinge pin support bracket comprises a bracket fastener aperture, wherein the base fastener aperture is aligned with the bracket fastener aperture when the hinge pin support bracket is secured relative to the base by the mechanical fastener.

5. A hinge assembly according to claim 4, wherein the hinge pin support bracket and the base of the track assembly comprise interlocking mechanical features configured to transfer mechanical loads directed along the track axis on the hinge pin to the base and the hinge pin support bracket when the hinge pin support bracket is secured relative to the base by the mechanical fastener.

6. A hinge assembly according to claim 5, wherein the interlocking mechanical features are located proximate the base fastener opening and the bracket fastener opening.

7. A hinge assembly according to 4, wherein the bracket fastener opening comprises an elongated non-circular opening such that a position of the hinge pin bracket relative to the base is adjustable along the track axis.

8. A hinge assembly according to claim 2, wherein the hinge pin support bracket comprises a vent limiter configured to limit rotation of the connecting arm around the hinge pin when the hinge assembly is moving from a closed configuration to an open configuration.

9. A hinge assembly according to claim 8, wherein the hinge pin support bracket and the vent limiter each comprise separate and discrete articles attached to each other.

10. A hinge assembly according to claim 1, wherein the hinge pin support comprises a hinge pin support bracket separate and discrete from the base of the track assembly, wherein a base portion of the hinge pin support bracket is positioned on the base and a support portion of the hinge pin support bracket is positioned above the base of the track and the base portion of the hinge pin support bracket, and wherein the base portion comprises an access aperture through which the hinge pin extends such that the hinge pin extends between the base and the support portion of the hinge pin support bracket.

11. A hinge assembly according to claim 10, wherein the support portion of the hinge pin support bracket comprises a hinge pin bore, wherein a portion of the hinge pin is located in the hinge pin bore such that movement of the hinge pin relative to the support portion of the hinge pin support bracket is restricted.

12. A hinge assembly according to claim 10, wherein the base portion of the hinge pin support bracket extends between a shoe stop end and a hinge pin end, wherein the access aperture is located closer to the hinge pin end than the shoe stop end, and wherein the shoe stop end is configured to prevent movement of the shoe past the shoe stop end towards the hinge pin.

13. A hinge assembly according to claim 10, wherein the hinge pin support bracket is secured relative to the base by a mechanical fastener.

14. A method of protecting a hinge pin of a window hinge assembly, the method comprising: attaching a base of a track assembly to a window frame, the track assembly comprising a shoe track, wherein the base extends along a track axis from a hinge pin end to a shoe track end, and wherein the shoe track extends along the track axis, and wherein the shoe track extends from the shoe track end towards the hinge pin end, attaching a base end of a hinge pin to the base proximate the hinge pin end of the base, the hinge pin comprising an upper end distal from the base; engaging a shoe engaged with the shoe track, wherein the shoe is configured to slide along the shoe track end the shoe is restrained from moving transverse to the track axis when the shoe is slidably engaged with the shoe track; attaching a sash arm to a rotating sash that is attached to the window frame, the sash arm extending from a shoe end to a distal end, wherein the shoe end is pivotally coupled to the shoe; pivotally connecting a sash end of a connecting arm to the sash arm at an intermediate

location spaced from the shoe end of the sash arm and pivotally connecting a hinge pin end of the connecting arm to the hinge pin, wherein the connecting arm rotates about a pin axis extending through the base end and the upper end of the hinge pin when the shoe moves along the shoe track and the sash arm is connected to the connecting arm; and restraining the upper end of the hinge pin above the base of the track assembly from movement relative to the base by locating the upper end of the hinge pin in a hinge pin bore located in a support portion of a hinge pin support attached to the base, wherein the support portion extends above the base, wherein pivotally connecting the connecting arm to the hinge pin comprises positioning the hinge pin in an opening in the connecting arm proximate the hinge pin end of connecting arm such that the hinge pin end of the connecting arm is located between the support portion of the hinge pin support and the base of the track assembly.

15. A method according to claim 14, wherein the hinge pin support comprises a hinge pin support bracket separate and discrete from the base of the track assembly, wherein the method comprises attaching the hinge pin support bracket to the base.